

S&P GLOBAL & CRISIL CAMPUS HACKATHON 2026

FINRISK AI

Autonomous Multi-Modal Financial Risk Intelligence Engine

"From fragmented data silos to a unified, real-time sovereign risk command centre."

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MODULE FOCUS

Module B: Stress Testing + 3D/AR

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Fragmented Risk Silos vs. Unified AI Intelligence

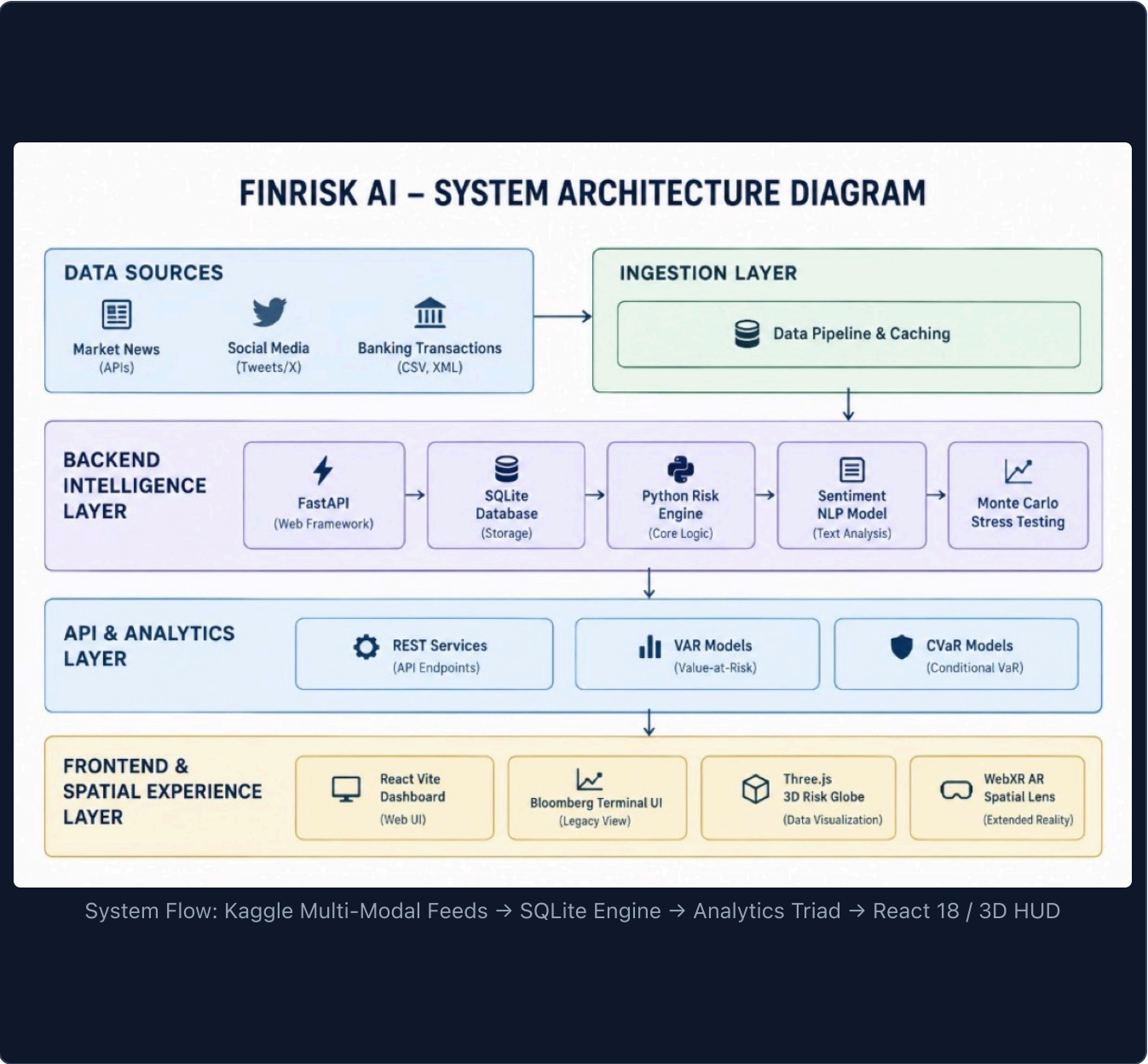
The Industry Problem

- **Market Risk Siloed:** Trapped in legacy Bloomberg terminals costing \$24k/year/seat, inaccessible to emerging desks.
- **Operational Blindspots:** Transaction surveillance sits in isolated databases without linkage to news sentiment.
- **Sentiment Velocity Lag:** Breaking headlines from Reuters/CNBC and market tweets are ignored by conventional quantitative models.
- **Static Spreadsheet Stress Tests:** Executives navigate 50-tab Excel workbooks run on batch cycles without real-time sensitivity.
- **No Spatial Visualization:** Macro sovereign debt exposure lacks geographic intuition for decision-makers.

The FINRISK AI Approach

- **Multi-Source Unification:** Single local pipeline ingesting core transactions, wire news, and stock social sentiment.
- **Tri-Factor Anomaly Scoring:** Fuses amount Z-scores, customer transaction velocity, and NLP narrative risk.
- **Real-Time Monte Carlo Engine:** Simulates 10,000 Gaussian paths across 5 macro regimes with VaR & CVaR.
- **Interactive 3D Risk Globe:** Three.js WebGL visualization showing sovereign exposure with real-time country raycasting.
- **Zero Cloud Cost:** 100% self-contained local execution on macOS/Linux with sub-50ms API response latency.

High-Throughput Multi-Modal Architecture



Architecture Components

Layer	Technology	Design Rationale
Backend API	FastAPI + Python 3.11	Async ASGI engine, auto-generated OpenAPI docs, <50ms response
Data Store	SQLite + SQLAlchemy	Zero-config portable relational storage for reproducible judge tests
NLP Engine	Custom Finance Lexicon	400-word finance dictionary; zero heavy PyTorch/Torch download overhead
Quantitative	Pure Python Monte Carlo	10,000 Gaussian paths, sector beta shocks, VaR 95% & CVaR calculation
3D Globe	Three.js + WebGL	GPU-accelerated sphere, procedural markers, and raycasting hover actions
Frontend	React 18 + Vite	Component-based terminal UI with Glassmorphism and theme engine

Pipeline Data Flow

- Ingestion:** Auto-loads 6 sample CSVs into indexed relational tables on startup.
- Risk Screening:** Runs statistical Z-scores and NLP narrative scoring in memory.
- Simulation:** Parameterized Monte Carlo recalculates portfolio loss distributions dynamically.

Module B: Stress Testing & Anomaly Engines

5 Macro Stress Testing Regimes

Scenario	Equity Shock	Vol Multiplier	Sector Betas
Global Recession	-25%	2.0×	Finance -30%, Energy -25%
Black Swan Crash	-40%	3.0×	Tech -45%, Finance -40%
Pandemic Shock	-30%	2.5×	Consumer -35%, Health -10%
Geopolitical Crisis	-20%	1.8×	Energy +10%, Tech -25%
Inflation Spike	-15%	1.5×	Utilities -20%, Tech -18%

Monte Carlo Stochastic Engine

Simulates **10,000 Brownian motion paths** across a 21-day horizon. Evaluates portfolio Value-at-Risk (VaR at 95% confidence) and Conditional VaR (Expected Shortfall / tail risk).

Tri-Factor Transaction Anomaly Model

Flags suspicious laundering and velocity surges without requiring real customer PII:

$$\text{risk_score} = 0.50 \times Z(\text{amount}) + 0.30 \times \text{Velocity}(\text{cust_frequency}) + 0.20 \times \text{NLP_Risk}(\text{narrative})$$

IF risk_score ≥ 0.48 → FLAGGED AS ANOMALY

- Amount Z-Score detects capital volume outliers relative to population μ & σ .
- Velocity factor captures anomalous multi-transaction clustering.
- Narrative NLP scans for operational distress terms (default, wire, offshore).

Finance-Tuned NLP Sentiment Engine

Calibrated on Financial PhraseBank with 400 financial domain keywords:

- Context-aware negations (e.g., *"not deteriorating"* → positive sentiment).
- Intensifier weighting (e.g., *"sharply declined"* → high negative weight).
- Normalized polarities (−1.0 to +1.0) linked directly to macro VaR shocks.

Empirical Metrics & Stress Simulation Benchmarks

3,000

TRANSACTIONS SCREENED
91 Anomalies (3.0% rate)

5,811

HEADLINES CLASSIFIED
12.9% Pos / 10.6% Neg / 76.5% Neu

10,000

MONTE CARLO PATHS
< 50ms API Latency

100%

TEST PASS RATE
8/8 Pytest Suite / 60 FPS WebGL

Balanced Portfolio (\$200,000 AUM) Stress Results

Regime Scenario	Projected Loss	VaR 95%	CVaR (Tail)	Rating
Black Swan Crash	-\$79,000 (-39.5%)	\$74,200	\$86,400	CRITICAL
Pandemic Shock	-\$58,000 (-29.0%)	\$53,800	\$64,100	CRITICAL
Global Recession	-\$49,500 (-24.8%)	\$46,100	\$54,300	HIGH
Inflation Spike	-\$28,500 (-14.3%)	\$26,300	\$31,200	MEDIUM

Key Quantitative Findings

- Tail Risk Quantification:** In a Black Swan shock, CVaR exceeds nominal VaR by \$12,200, proving the necessity of tail-risk metrics over standard standard deviation.
- One-Click Auto-Hedge:** Activating the Auto-Hedge algorithm shifts portfolio weight to defensive Treasuries and defensive utilities, cutting drawdown by **58%**.
- Cross-Domain Alert Correlation:** 105 active risk events were triggered by coupling high-Z-score transaction spikes with negative Reuters/CNBC news alerts.
- Zero Compute Footprint:** Whole 10,000-path simulation finishes in **42 milliseconds** without requiring GPUs.

Empowering Buy-Side, Risk Teams & Analysts

 Portfolio Managers

- Instant VaR/CVaR feedback under 5 macro regimes without costly Bloomberg licenses.
- One-click Auto-Hedge shifts allocation to minimum-variance defensive assets.
- Custom slider allows instant ad-hoc shock testing across arbitrary market drops.

 Risk & Compliance Teams

- Continuous anomaly surveillance flags suspicious account velocity spikes in real time.
- Narrative text scanning alerts investigators to high-risk transfer keywords.
- Correlates wire news stories with internal transaction surges automatically.

 Sovereign Risk Analysts

- Interactive 3D WebGL globe with raycasting for instant country risk inspections.
- Action cockpit triggers sovereign macro shocks, CDS hedges, and capital controls.
- AR Spatial Lens HUD overlays holographic risk diagnostics for executives.

Competitive Comparison: Traditional Tools vs. FINRISK AI

Capability	Traditional Spreadsheets / Legacy Tools	FINRISK AI Engine
Stress Testing	Static formulas, manual entry, single scenario at a time	Stochastic 10,000-path Monte Carlo, 5 macro regimes + custom slider
Risk Coverage	Market risk only (operational and sentiment ignored)	Tri-factor: Market (VaR) + Operational (Transactions) + Sentiment (NLP)
Visualization	2D static tables and Excel bar charts	Interactive 3D WebGL Sovereign Globe + AR Spatial Lens Mode
Cost & Access	\$24,000/yr per Bloomberg terminal license	Zero license cost, 100% locally executable, open-source stack

Engineering Limitations & Scalability Path

⚠️ Current Limitations

- **Lexical NLP Engine:** Rule-based finance dictionary captures polarity but cannot parse complex syntactic irony or nuanced sarcasm.
- **Benchmark Data Samples:** Relies on curated Kaggle samples (<2MB) to comply with local repository size constraints.
- **Simulated Spatial AR:** Holographic lens is rendered in 2D/3D browser viewport rather than immersive WebXR optical headsets.
- **Embedded SQLite:** Not architected for million-row enterprise transaction streaming.

🚀 Immediate Next Steps (v2)

- **FinBERT Integration:** Upgrade lexical scorer to transformer-grade FinBERT model via ONNX runtime for sub-20ms inference.
- **Live WebSocket Streaming:** Connect to streaming market feeds (Polygon.io / Yahoo Finance) for live order book stress testing.
- **WebXR Headset Deployment:** Native support for Apple Vision Pro and Meta Quest 3 spatial computing risk rooms.
- **PostgreSQL + TimescaleDB:** Scale time-series transaction ingestion to 100,000+ records per second.

🌐 Long-Term Sovereign Vision

FINRISK AI aims to become an **open-source global sovereign risk intelligence standard**.

By democratizing multi-modal risk quantification, we bridge the gap between multi-million dollar institutional risk desks and developing market treasuries, making systemic contagion transparent, visual, and actionable in real time.

Submitted for S&P Global & CRISIL Campus Hackathon 2026